

Busca em profundidade p/ digrafos

Inicialização

$t \leftarrow 0$ -- t é o relógio ou tempo global

para todo vértice v em $V(G)$ faça

$PE(v) \leftarrow 0$ -- $PE(v)$ é a profundidade de entrada de v

$PS(v) \leftarrow 0$ -- $PS(v)$ é a profundidade de saída de v

$pai(v) \leftarrow \text{null}$ -- ponteiros que definem a floresta de profundidade T

Chamada Externa

enquanto existe v em $V(G)$ tal que $PE(v) = 0$ faça

executar $P(v)$ -- nova raiz da busca

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Procedimento recursivo de busca p/ digrafos

procedimento $P(v)$

$t \leftarrow t + 1; PE(v) \leftarrow t$

para todo vértice w em $N_{out}(v)$ faça

se $PE(w) = 0$ (*se w ainda não foi alcançado pela busca*)

então { marcar vw como *aresta “azul”* da floresta de profundidade T
 $pai(w) \leftarrow v$
executar $P(w)$

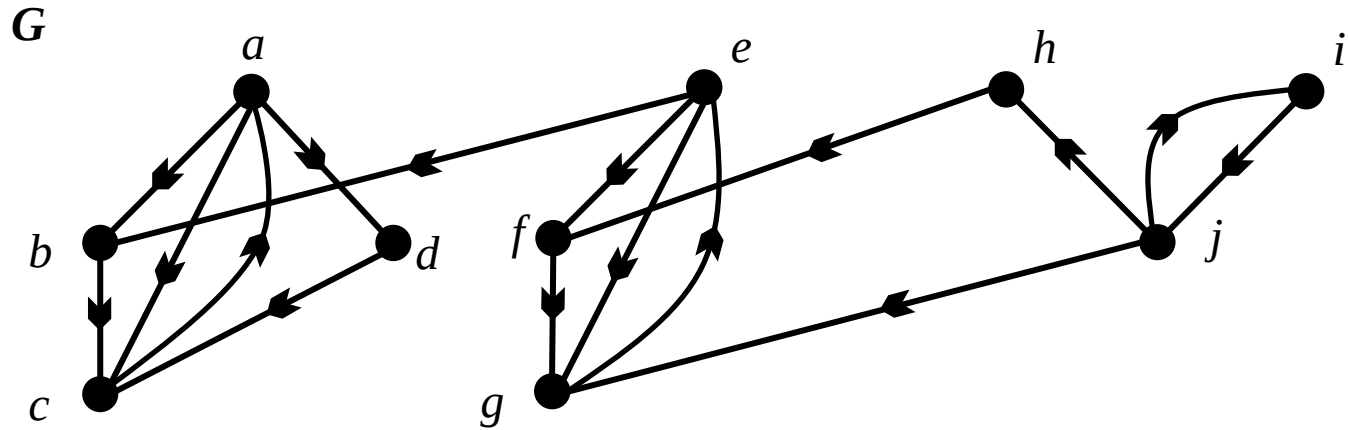
senão { se $PS(w) = 0$ (*se w ainda não saiu da busca*)
então marcar vw como *aresta “vermelha”* de retorno
senão se $PE(v) < PE(w)$ (*se v entrou antes de w na busca*)
então marcar vw como *aresta “amarela”* de avanço
senão marcar vw como *aresta “verde”* cruzamento

fim-para

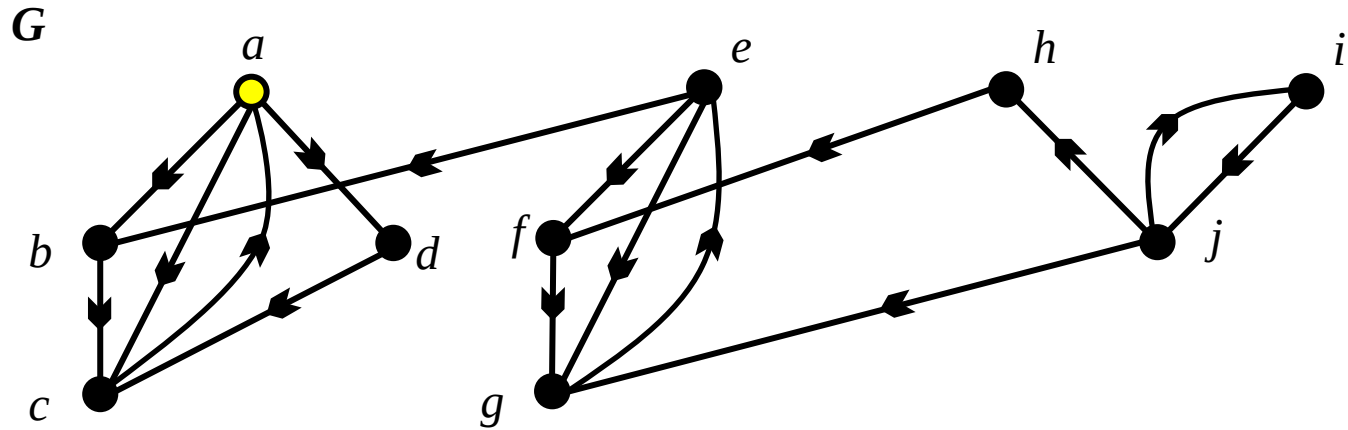
$t \leftarrow t + 1; PS(v) \leftarrow t$

fim-do-procedimento

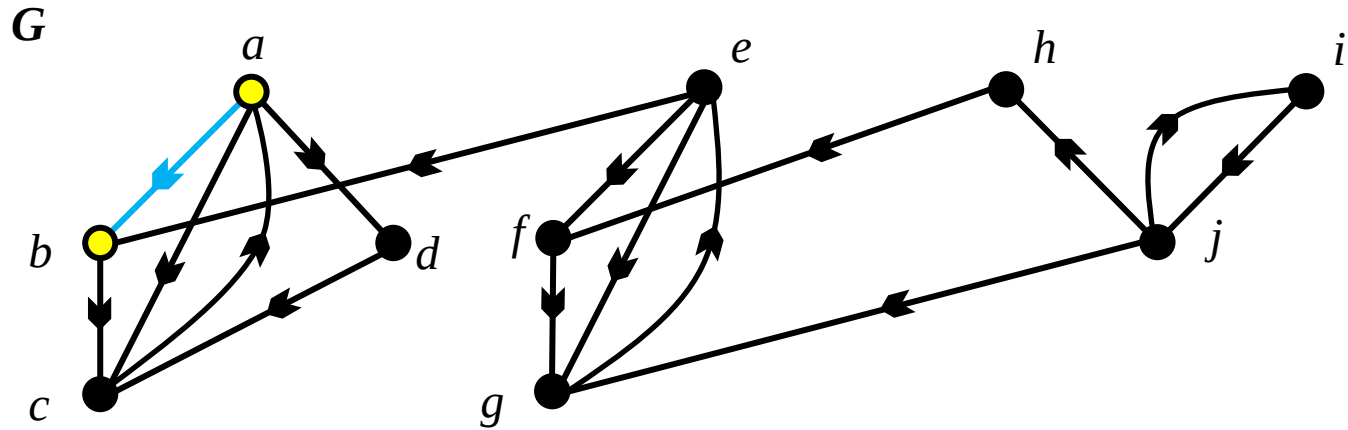
Busca em profundidade p/ digrafos

[illegible]

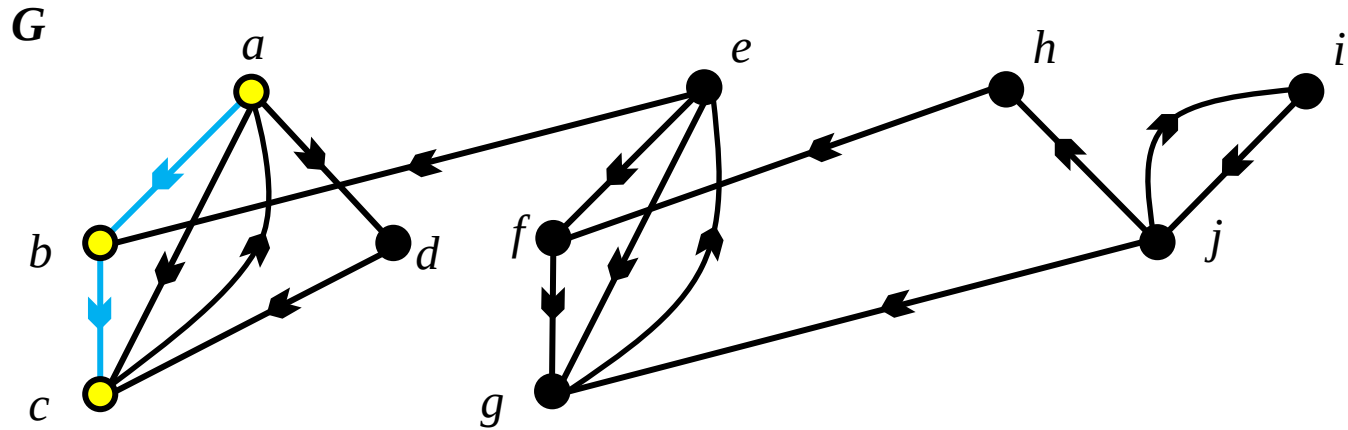
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[illegible]

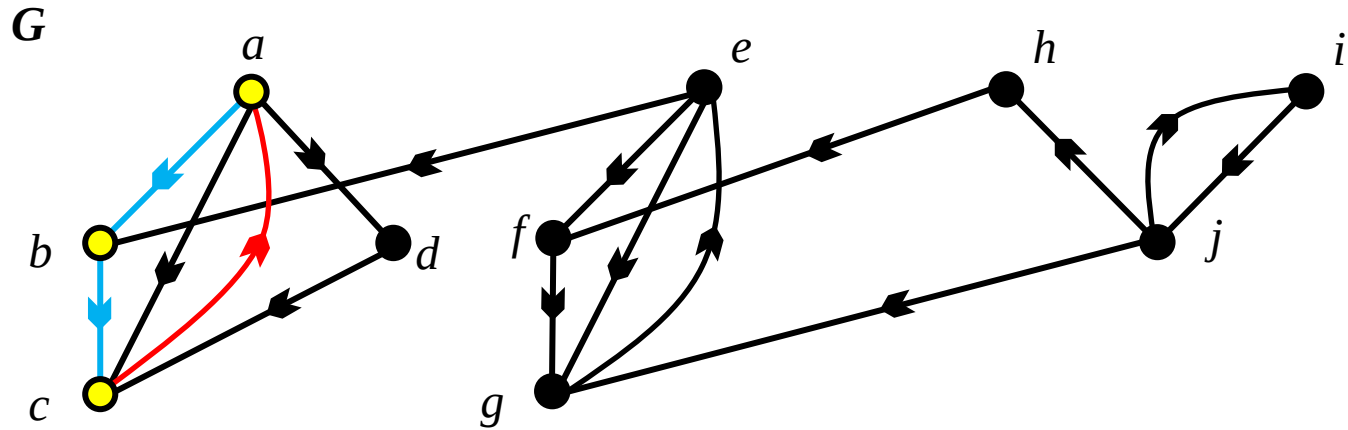
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[illegible]

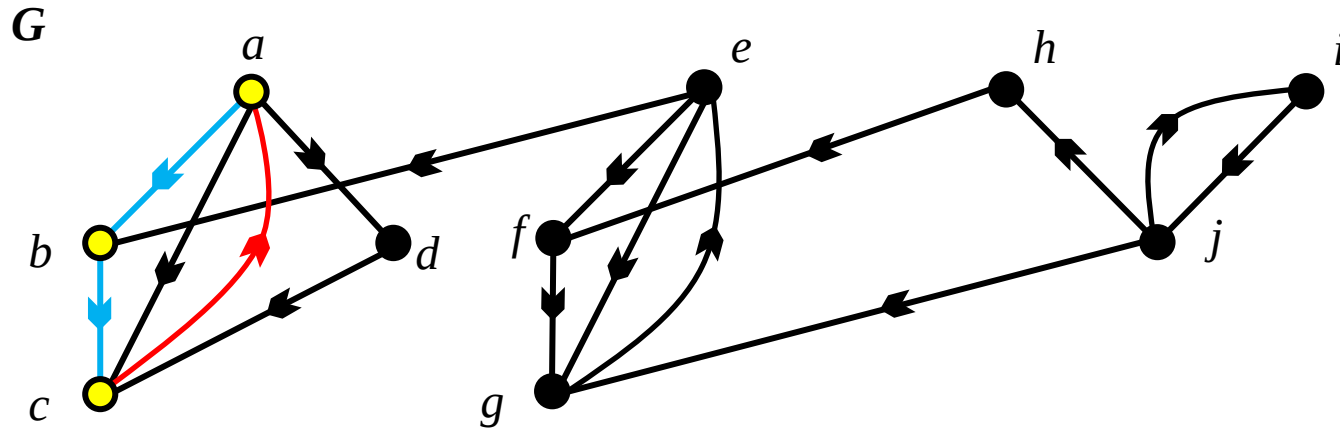
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[illegible]

Busca em profundidade p/ digrafos

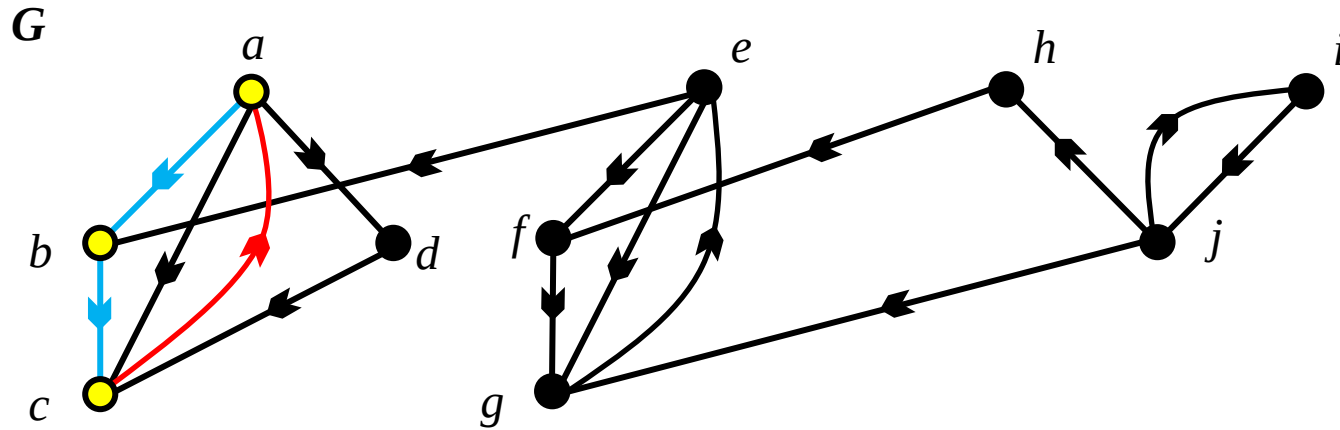
[illegible]

Busca em profundidade p/ digrafos



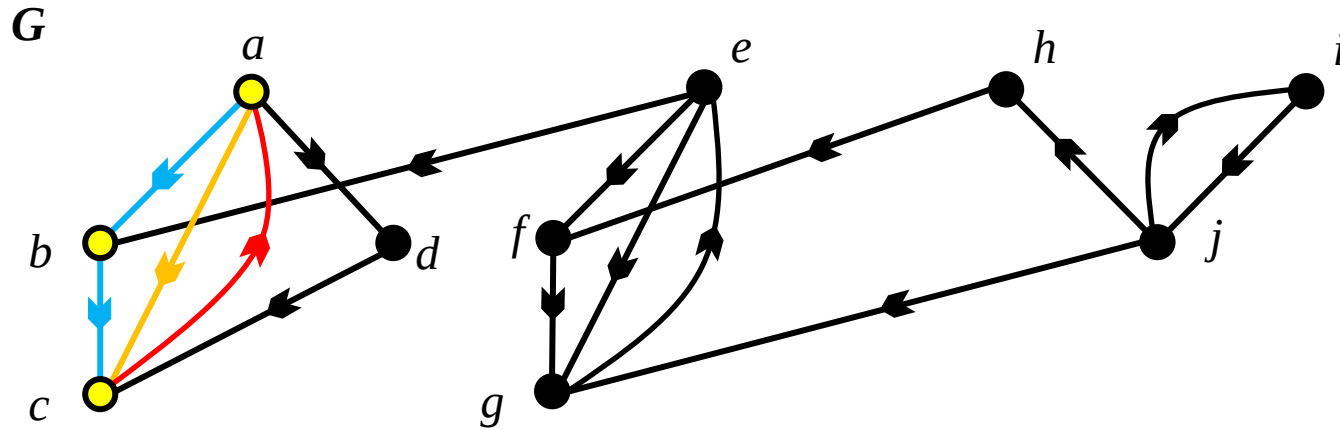
v	a	b	c	d	e	f	g	h	i	j
$PE(v)$	1	2	3	0	0	0	0	0	0	0
$PS(v)$	0	0	4	0	0	0	0	0	0	0

Busca em profundidade p/ digrafos



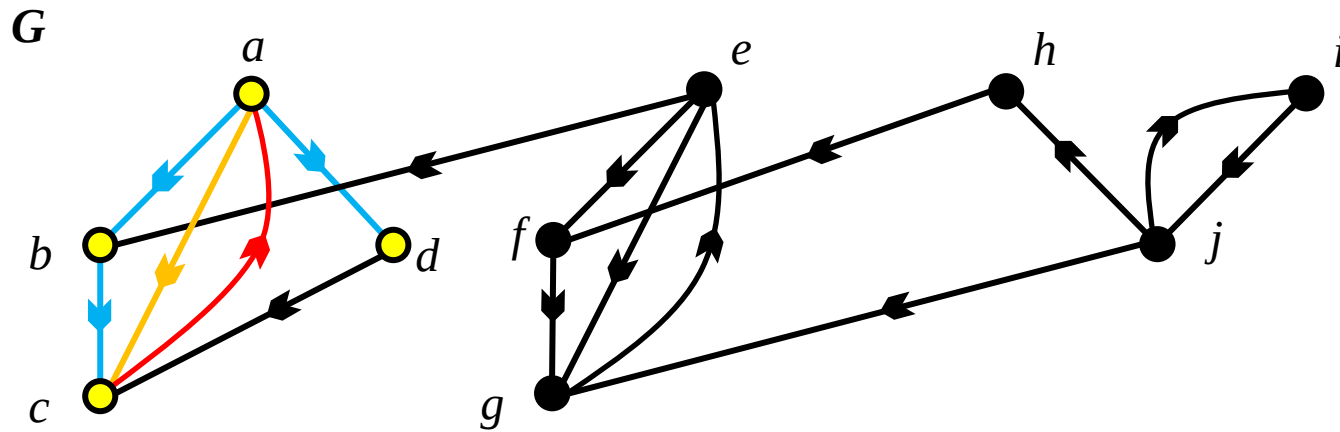
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE</i> (<i>v</i>)	1	2	3	0	0	0	0	0	0	0
<i>PS</i> (<i>v</i>)	0	5	4	0	0	0	0	0	0	0

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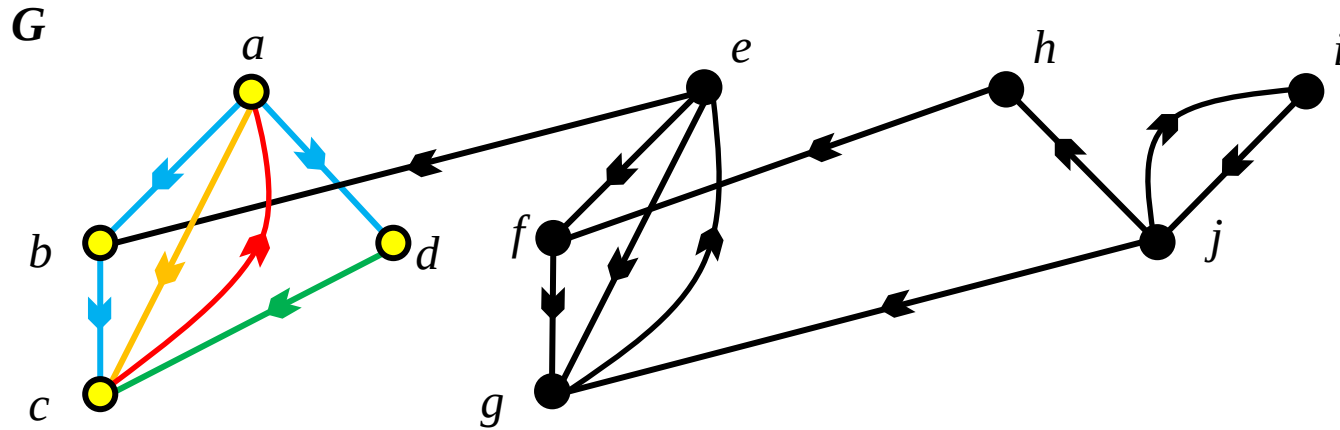
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	0	0	0	0	0	0	0
<i>PS(v)</i>	0	5	4	0	0	0	0	0	0	0

Busca em profundidade p/ digrafos



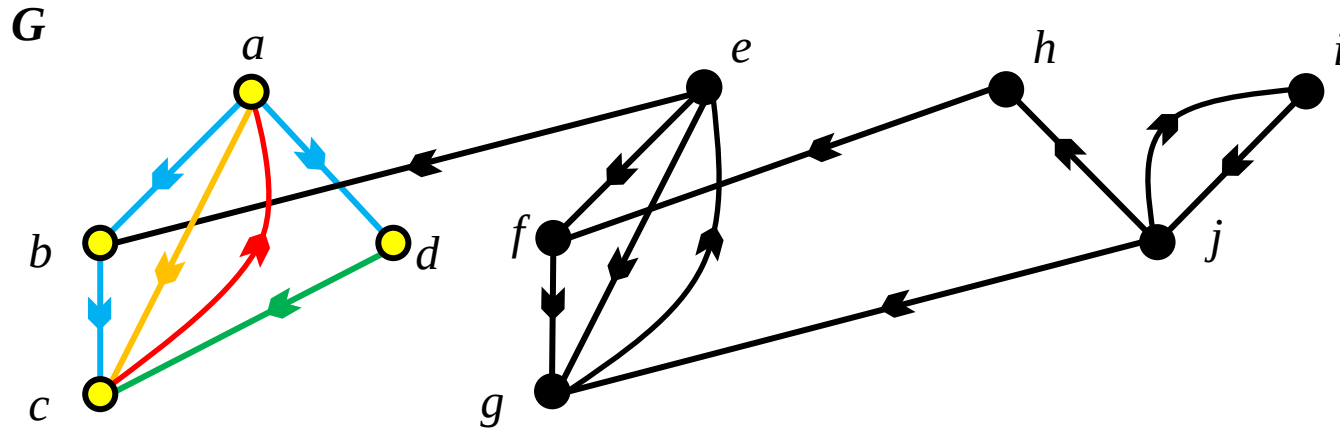
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	0	0	0	0	0	0
<i>PS(v)</i>	0	5	4	0	0	0	0	0	0	0

Busca em profundidade p/ digrafos



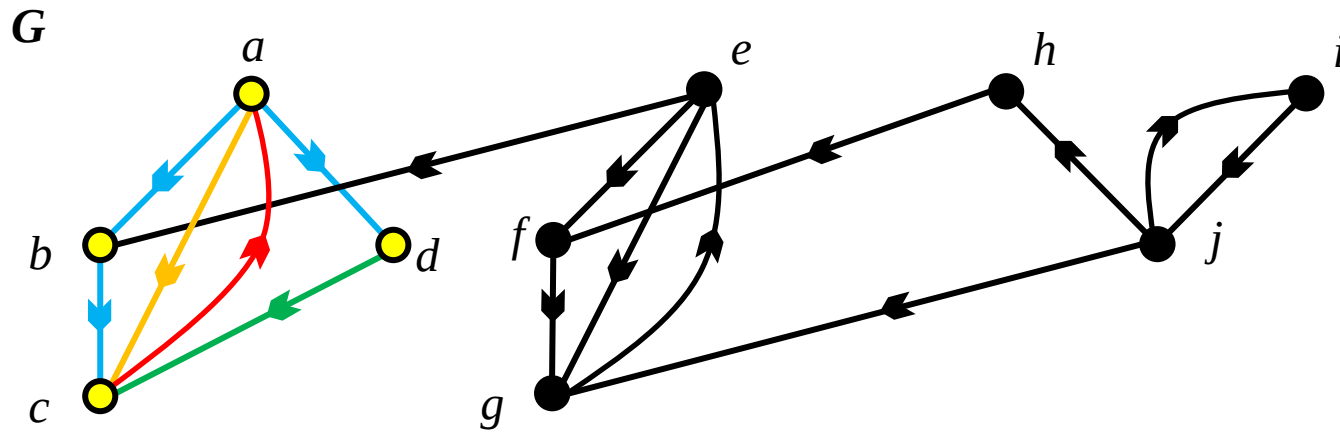
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	0	0	0	0	0	0
<i>PS(v)</i>	0	5	4	0	0	0	0	0	0	0

Busca em profundidade p/ digrafos



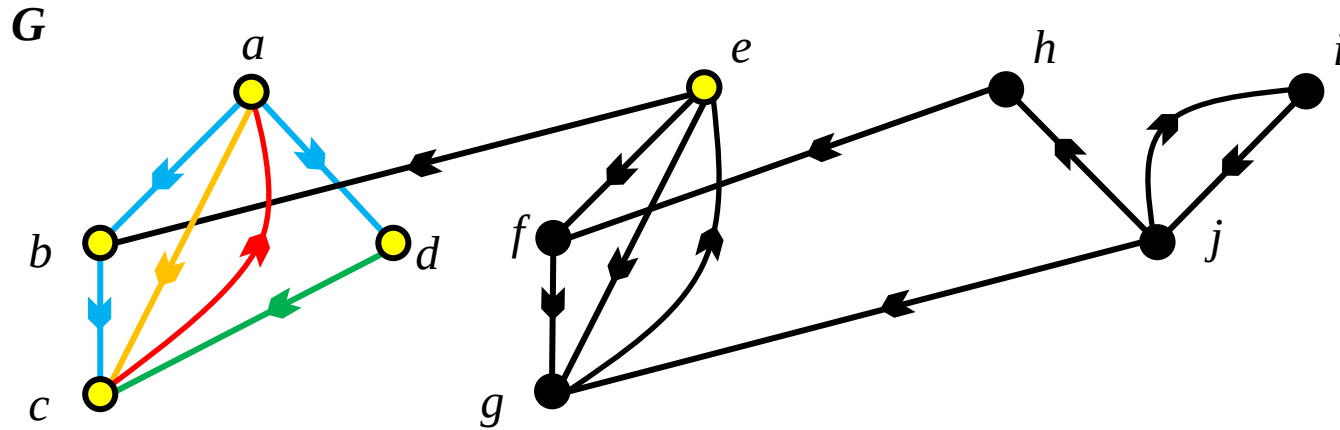
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	0	0	0	0	0	0
<i>PS(v)</i>	0	5	4	7	0	0	0	0	0	0

Busca em profundidade p/ digrafos



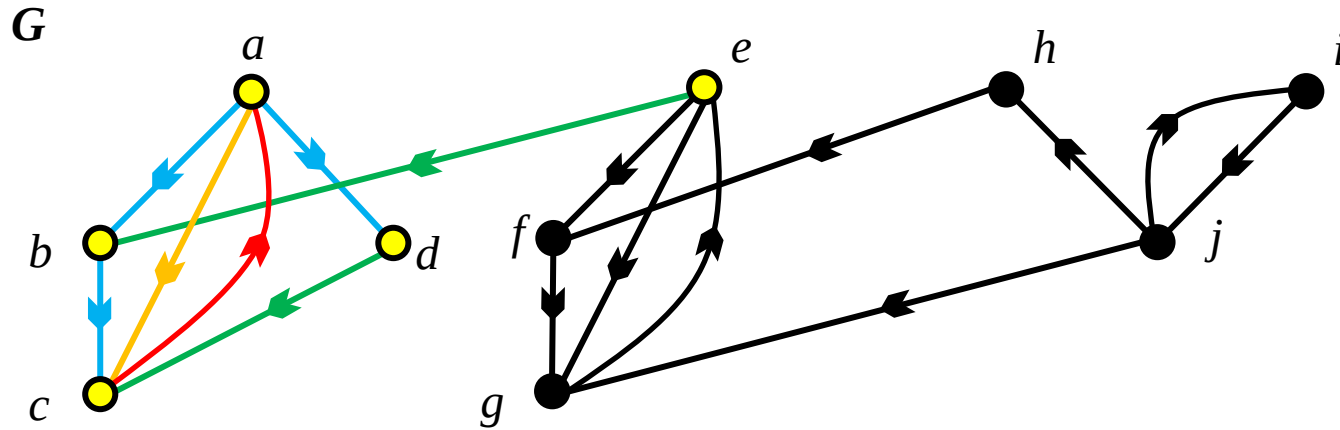
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE</i> (<i>v</i>)	1	2	3	6	0	0	0	0	0	0
<i>PS</i> (<i>v</i>)	8	5	4	7	0	0	0	0	0	0

Busca em profundidade p/ digrafos



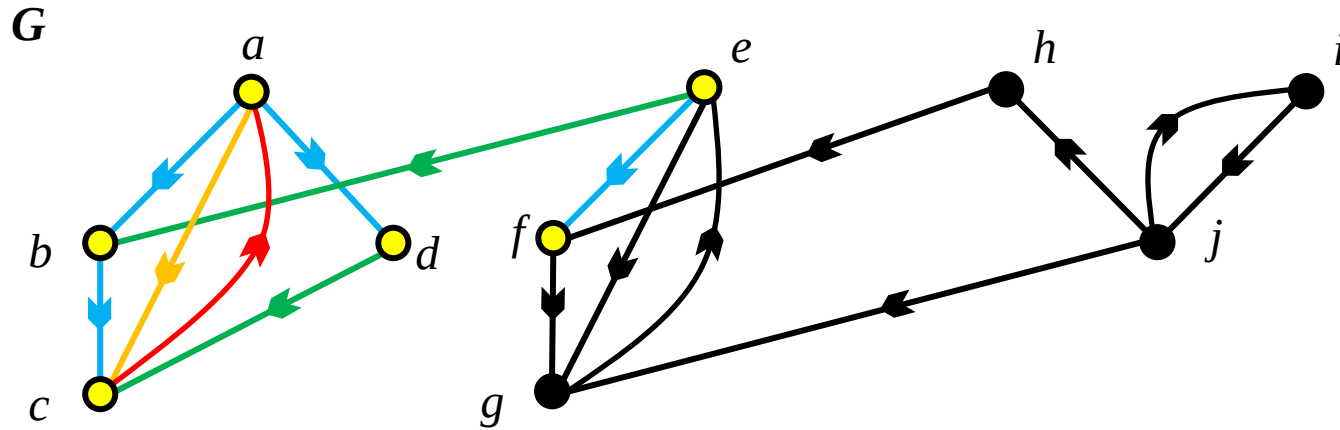
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	0	0	0	0	0
<i>PS(v)</i>	8	5	4	7	0	0	0	0	0	0

Busca em profundidade p/ digrafos



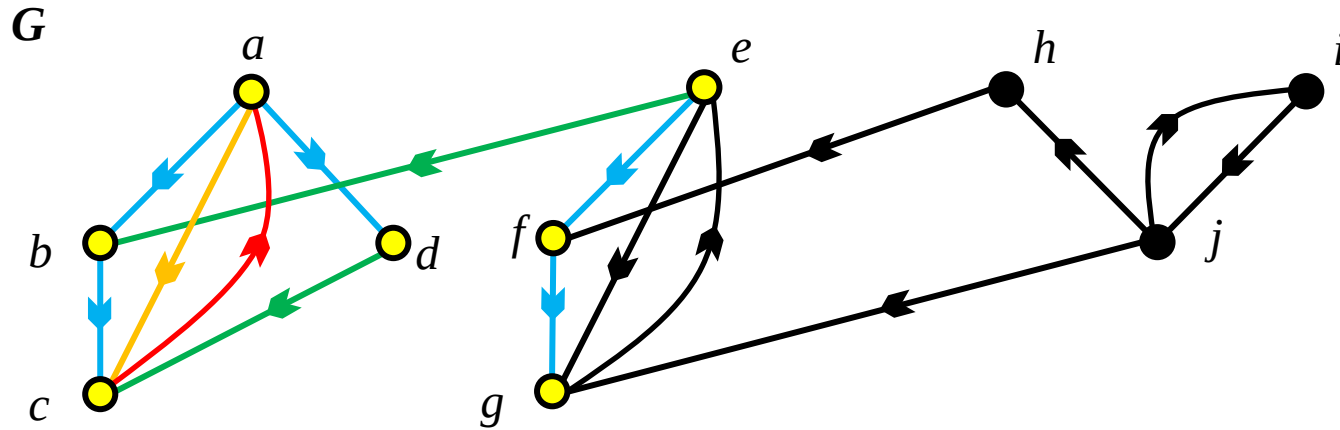
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	0	0	0	0	0
<i>PS(v)</i>	8	5	4	7	0	0	0	0	0	0

Busca em profundidade p/ digrafos



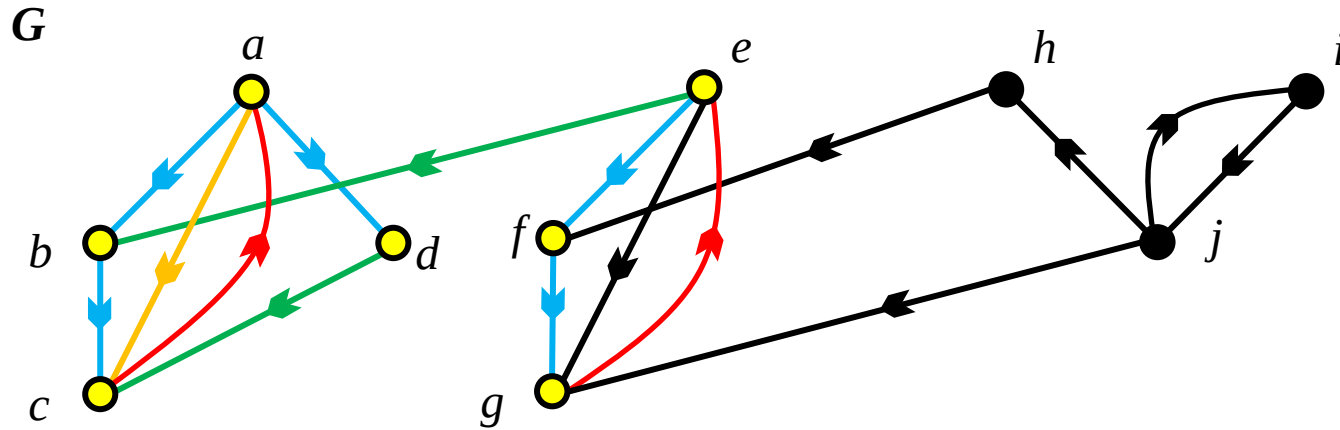
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE</i> (<i>v</i>)	1	2	3	6	9	10	0	0	0	0
<i>PS</i> (<i>v</i>)	8	5	4	7	0	0	0	0	0	0

Busca em profundidade p/ digrafos



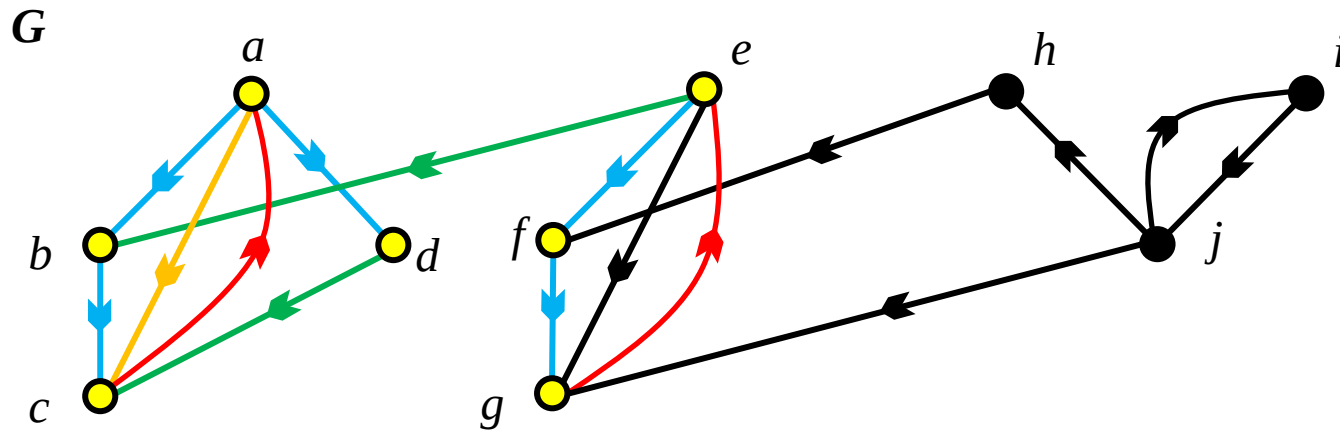
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	10	11	0	0	0
<i>PS(v)</i>	8	5	4	7	0	0	0	0	0	0

Busca em profundidade p/ digrafos



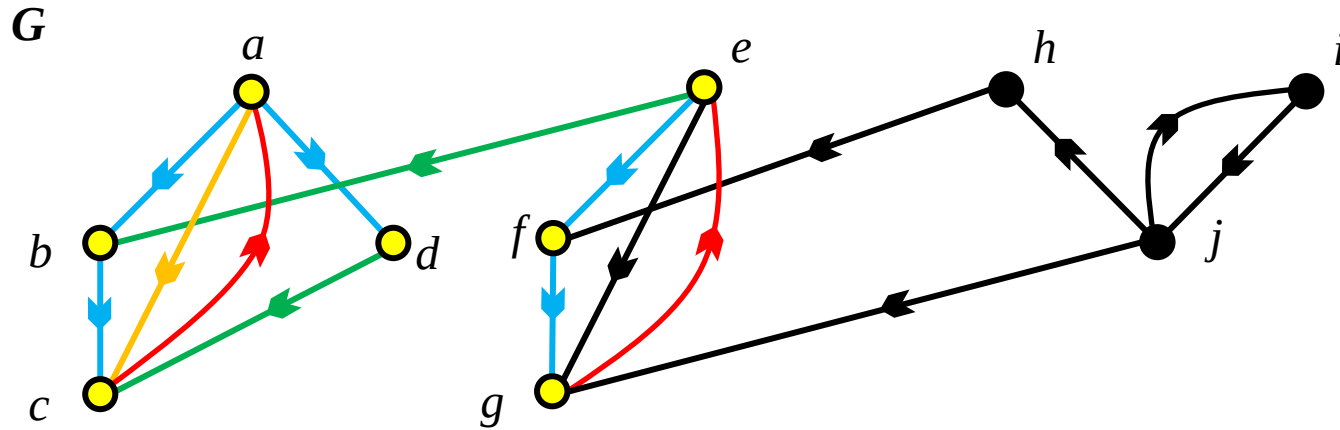
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	10	11	0	0	0
<i>PS(v)</i>	8	5	4	7	0	0	0	0	0	0

Busca em profundidade p/ digrafos



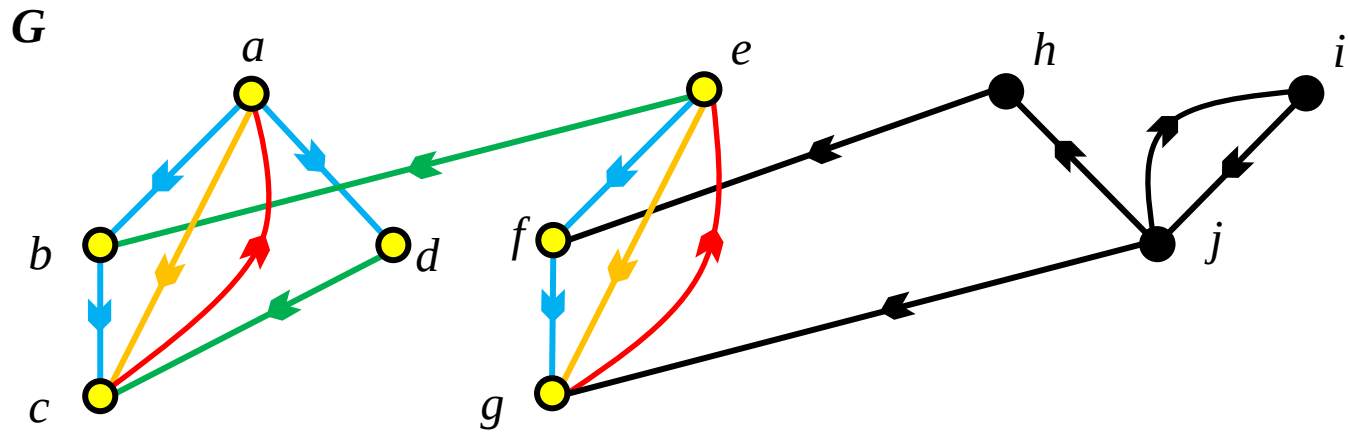
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	10	11	0	0	0
<i>PS(v)</i>	8	5	4	7	0	0	12	0	0	0

Busca em profundidade p/ digrafos



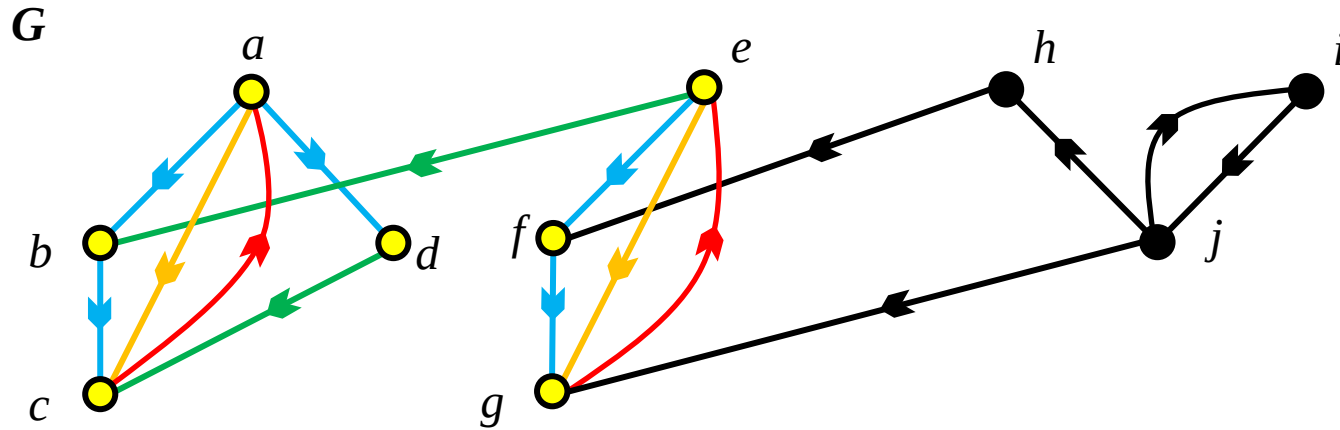
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE</i> (<i>v</i>)	1	2	3	6	9	10	11	0	0	0
<i>PS</i> (<i>v</i>)	8	5	4	7	0	13	12	0	0	0

Busca em profundidade p/ digrafos



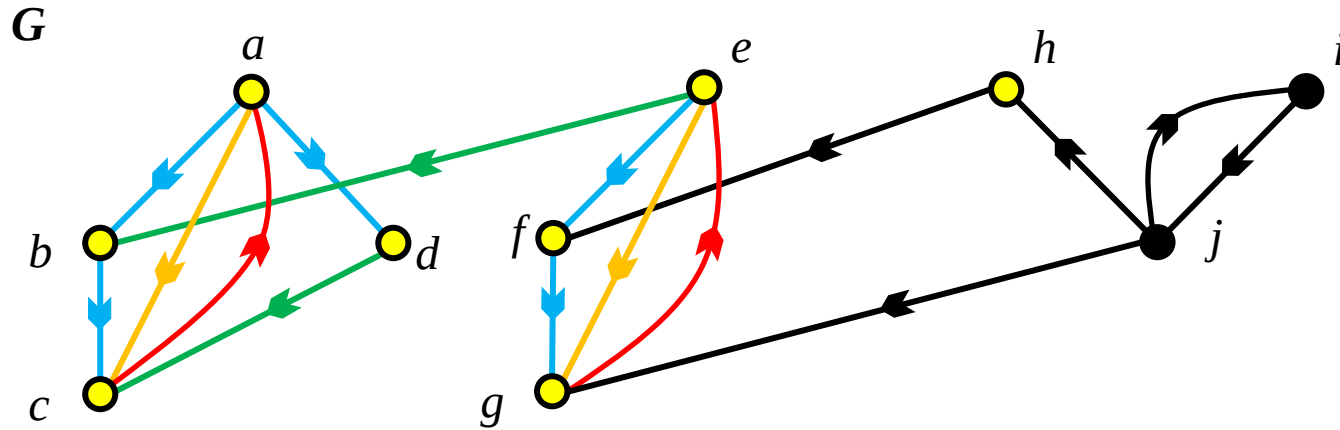
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	10	11	0	0	0
<i>PS(v)</i>	8	5	4	7	0	13	12	0	0	0

Busca em profundidade p/ digrafos



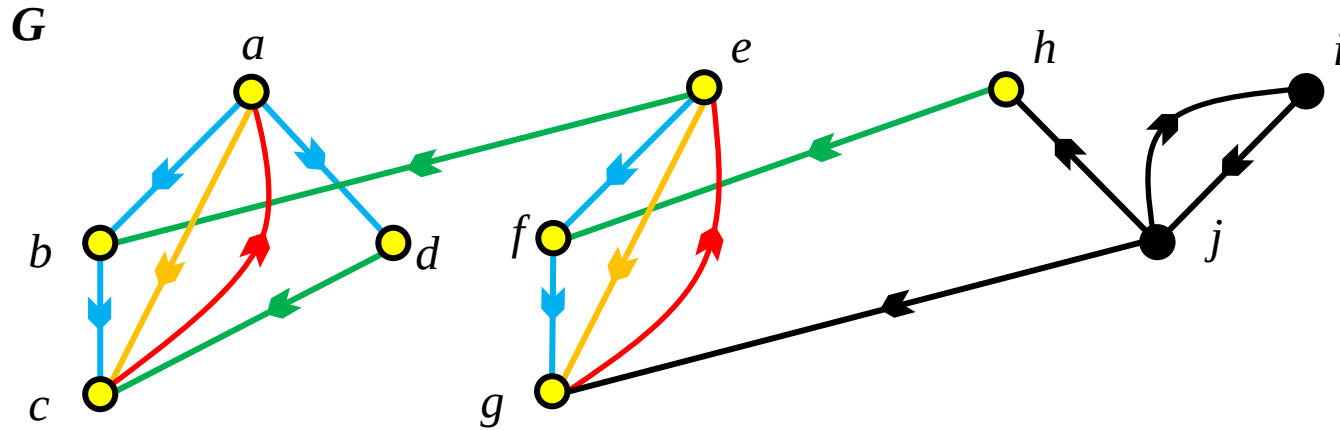
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	10	11	0	0	0
<i>PS(v)</i>	8	5	4	7	14	13	12	0	0	0

Busca em profundidade p/ digrafos



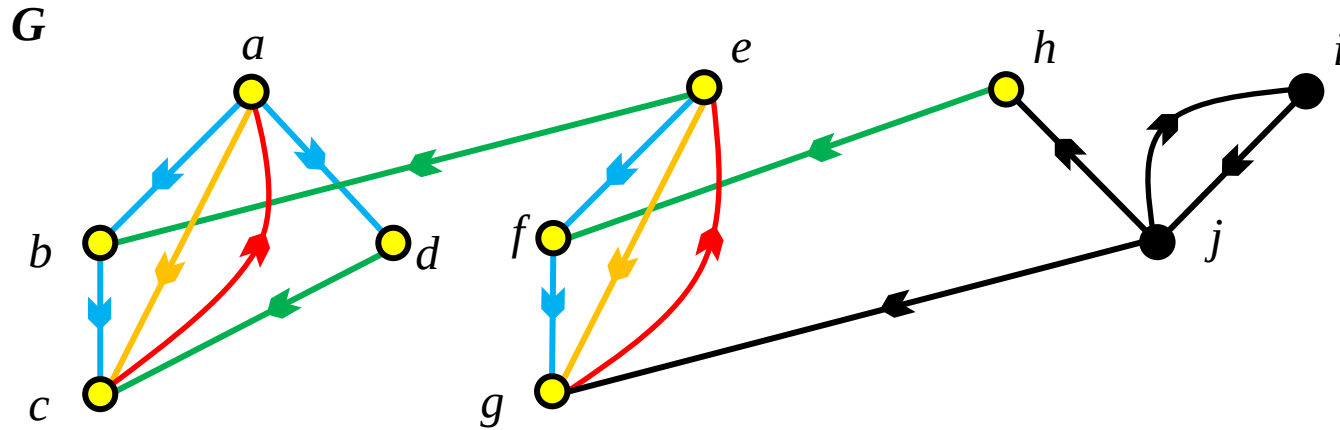
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE</i> (<i>v</i>)	1	2	3	6	9	10	11	15	0	0
<i>PS</i> (<i>v</i>)	8	5	4	7	14	13	12	0	0	0

Busca em profundidade p/ digrafos



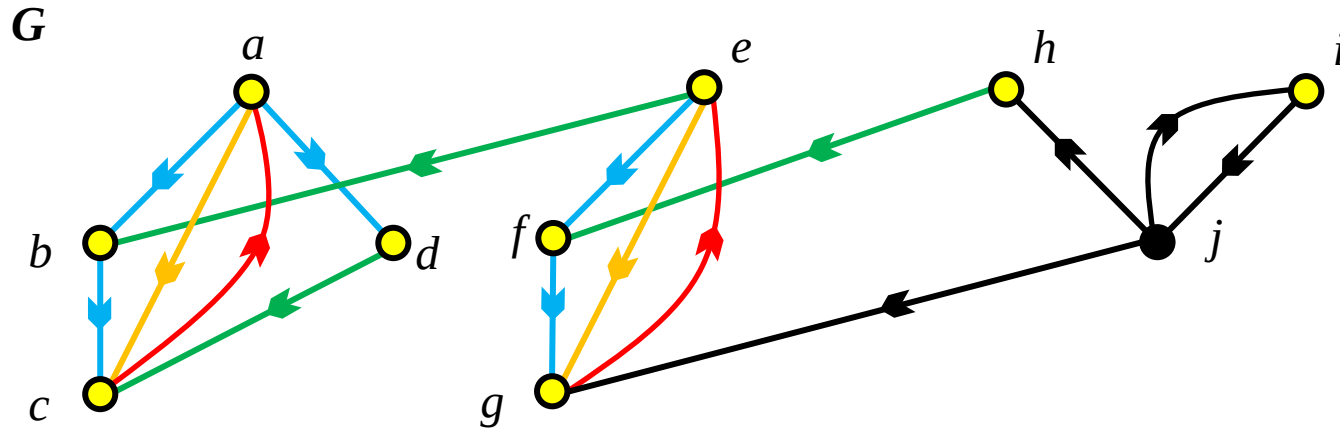
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	10	11	15	0	0
<i>PS(v)</i>	8	5	4	7	14	13	12	0	0	0

Busca em profundidade p/ digrafos



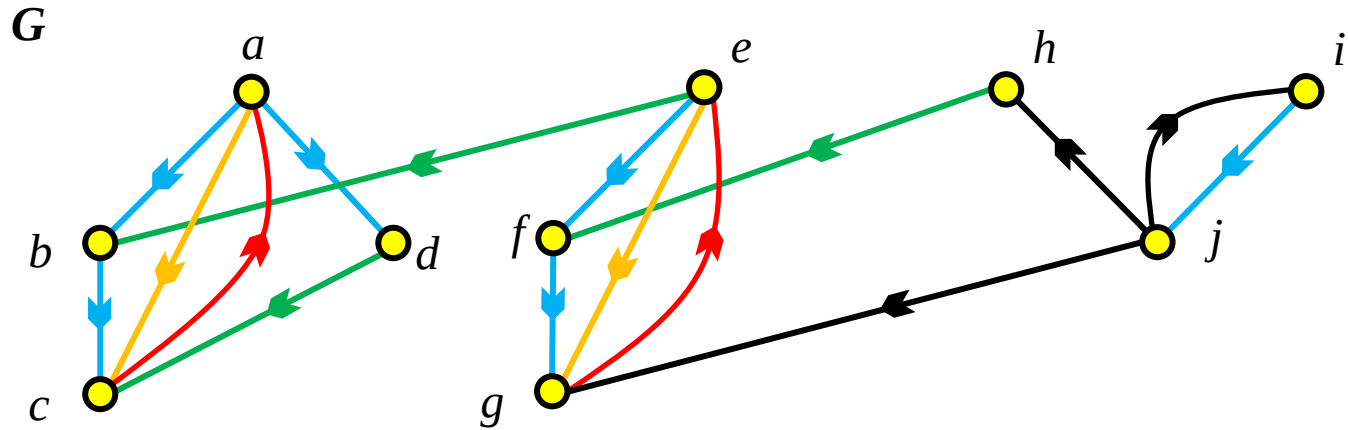
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	10	11	15	0	0
<i>PS(v)</i>	8	5	4	7	14	13	12	16	0	0

Busca em profundidade p/ digrafos



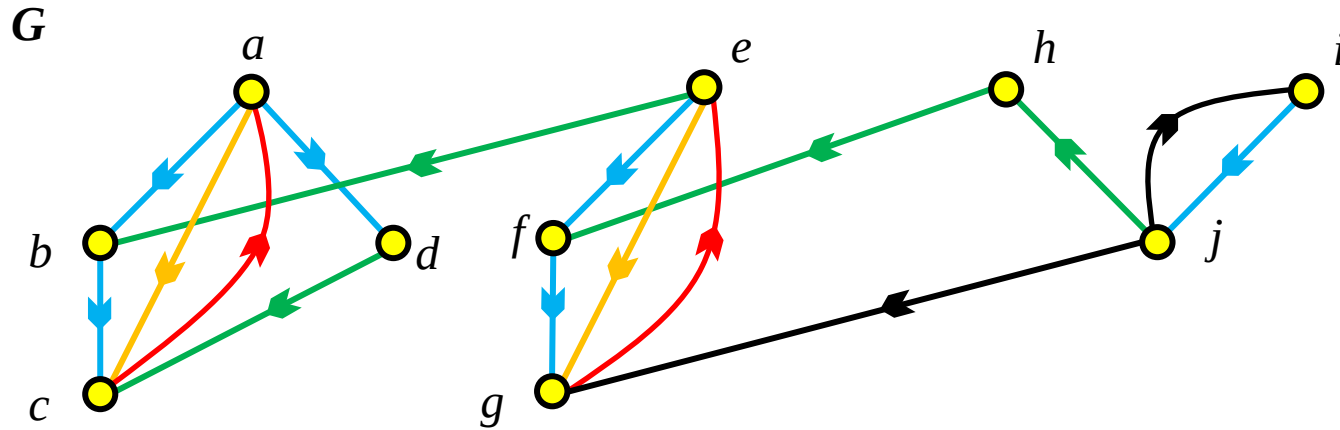
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	10	11	15	17	0
<i>PS(v)</i>	8	5	4	7	14	13	12	16	0	0

Busca em profundidade p/ digrafos



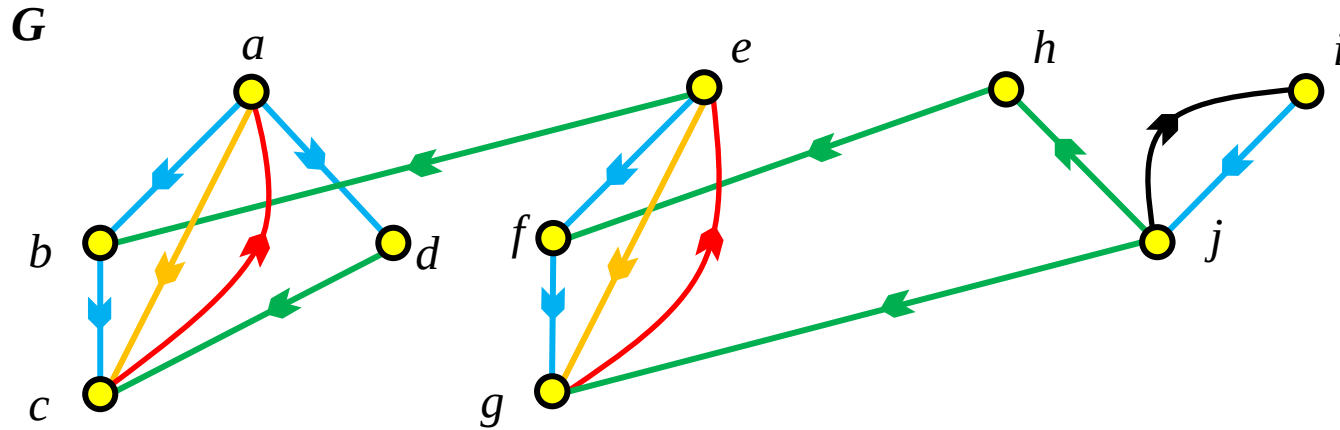
v	a	b	c	d	e	f	g	h	i	j
$PE(v)$	1	2	3	6	9	10	11	15	17	18
$PS(v)$	8	5	4	7	14	13	12	16	0	0

Busca em profundidade p/ digrafos



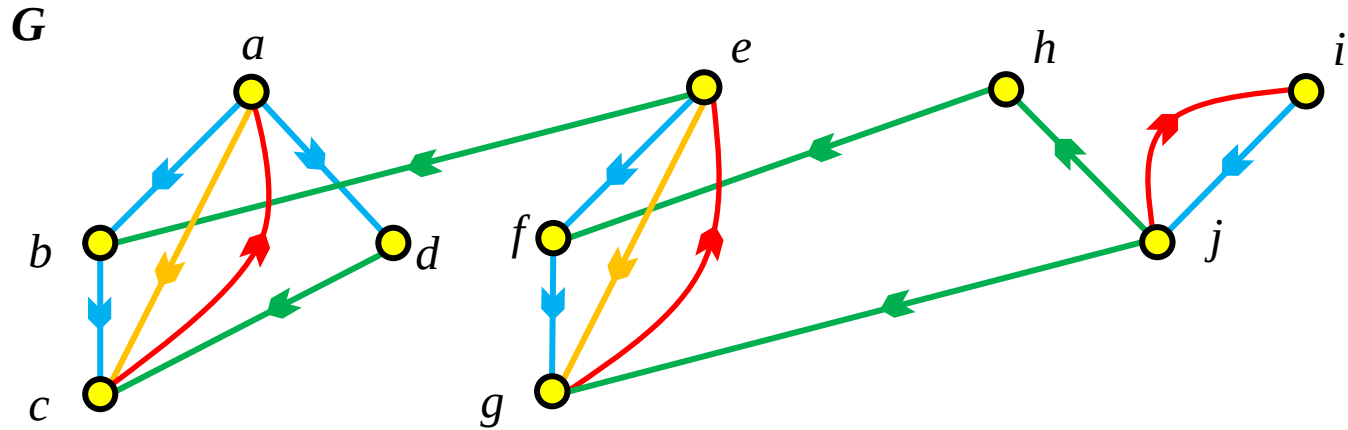
v	a	b	c	d	e	f	g	h	i	j
$PE(v)$	1	2	3	6	9	10	11	15	17	18
$PS(v)$	8	5	4	7	14	13	12	16	0	0

Busca em profundidade p/ digrafos



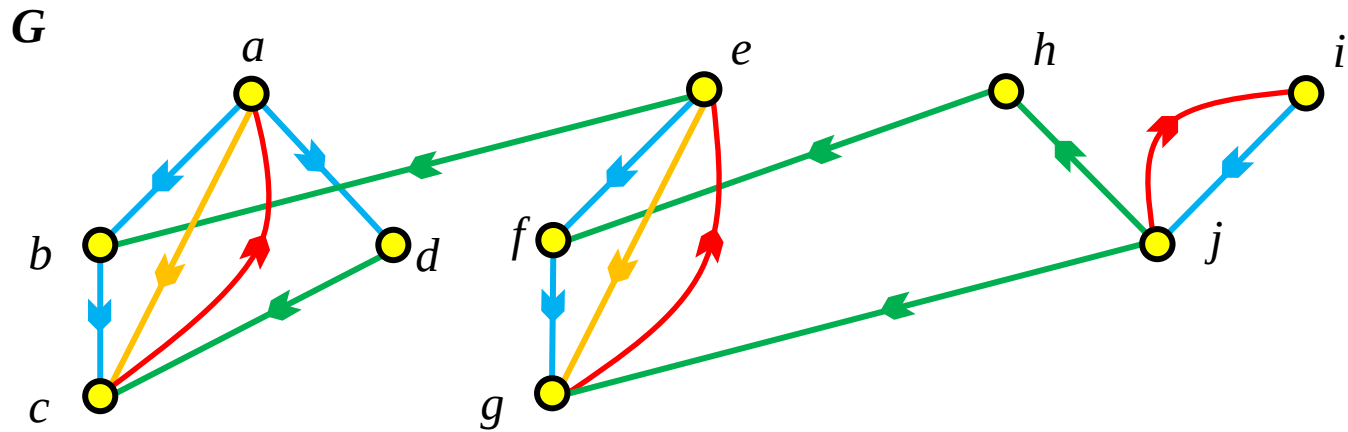
v	a	b	c	d	e	f	g	h	i	j
$PE(v)$	1	2	3	6	9	10	11	15	17	18
$PS(v)$	8	5	4	7	14	13	12	16	0	0

Busca em profundidade p/ digrafos



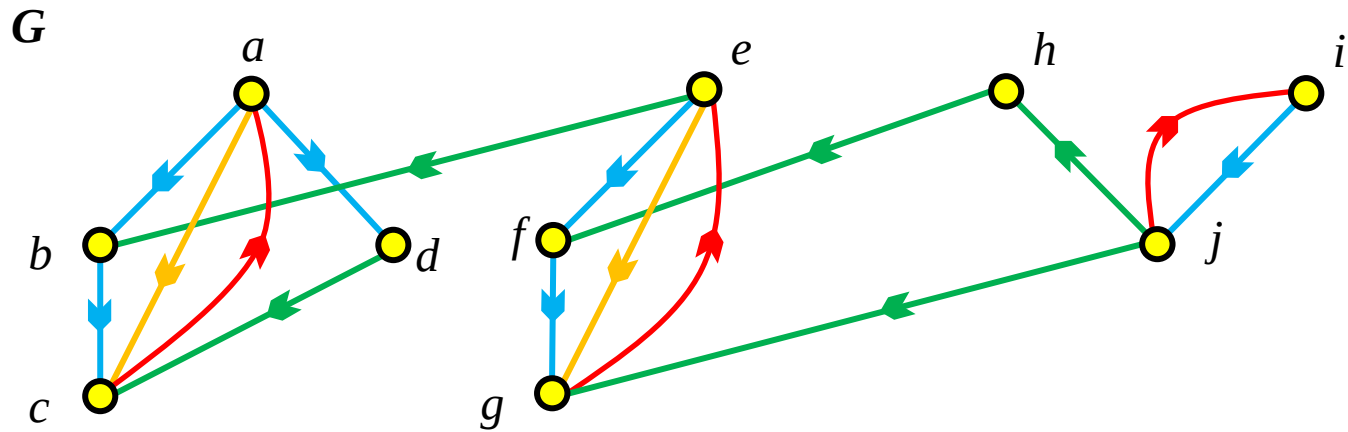
<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	10	11	15	17	18
<i>PS(v)</i>	8	5	4	7	14	13	12	16	0	0

Busca em profundidade p/ digrafos



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	10	11	15	17	18
<i>PS(v)</i>	8	5	4	7	14	13	12	16	0	19

Busca em profundidade p/ digrafos



<i>v</i>	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>
<i>PE(v)</i>	1	2	3	6	9	10	11	15	17	18
<i>PS(v)</i>	8	5	4	7	14	13	12	16	20	19